

HR Attrition Risk Analytics

*From Historical Data to
Predictive Retention*



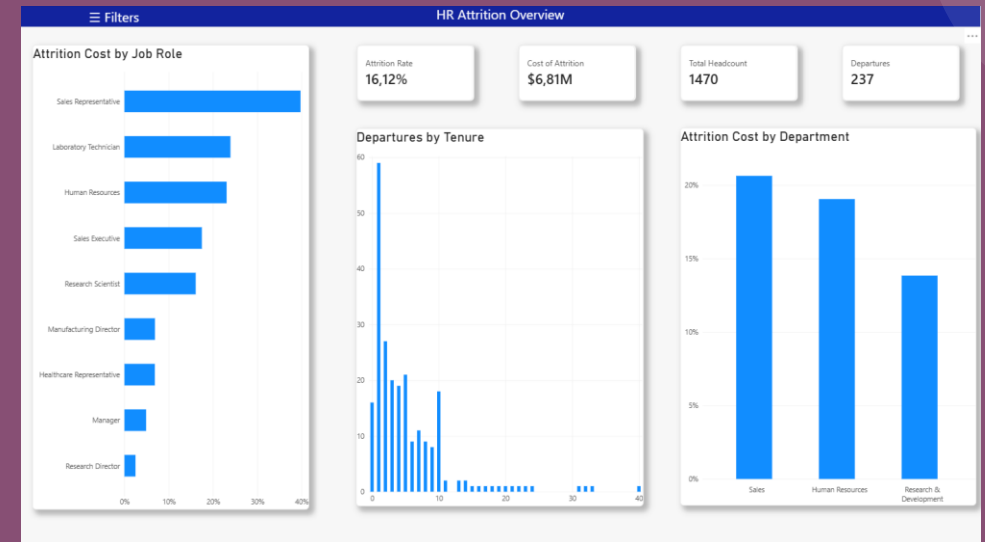


The Challenge: High costs of employee turnover and reactive Hr processes.

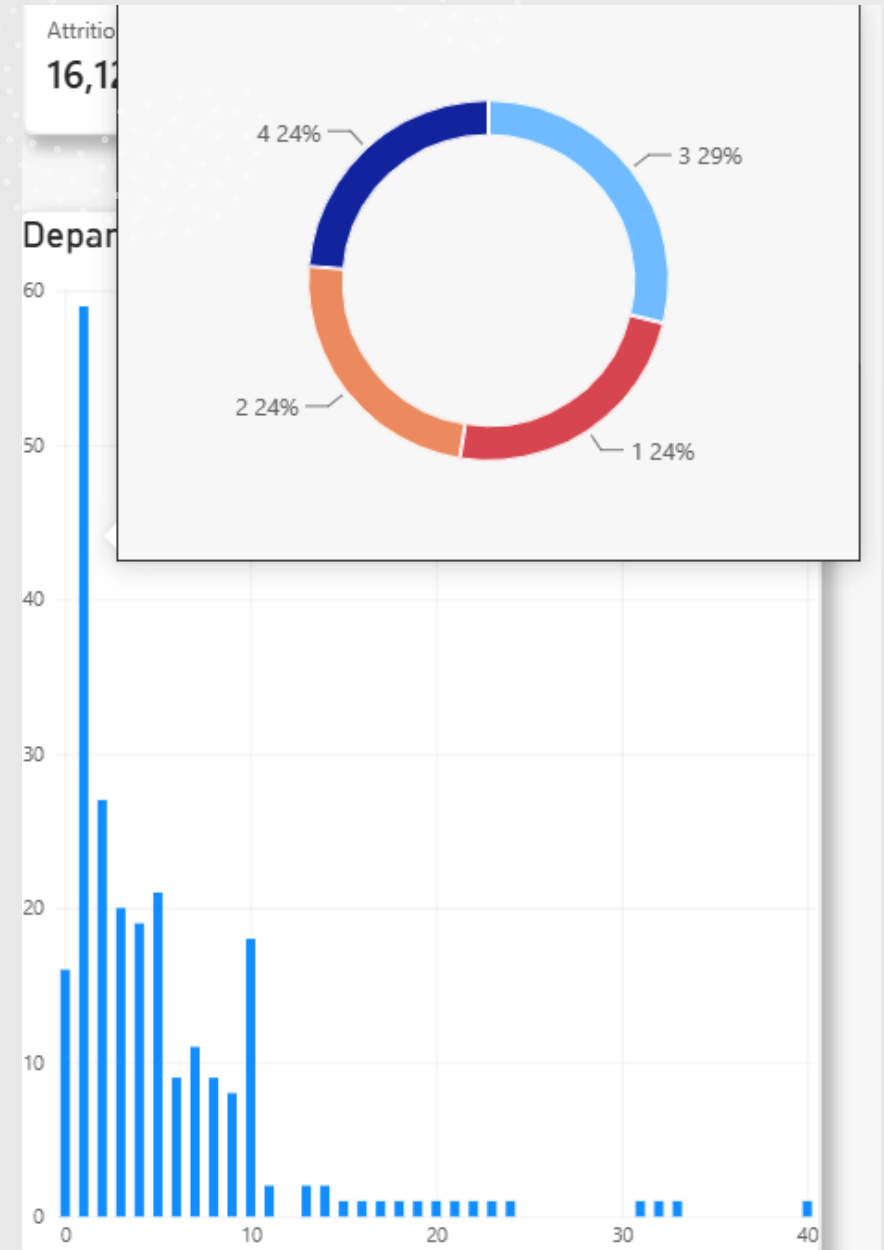


The Solution: An integrated analytics ecosystem (Power BI + Machine Learning) to monitor historical trends and predict future flight risks.

- *Current State:* 16.12% attrition rate generating \$6.81M in estimated turnover costs.
- *Key Insights:* 1. Sales Representatives generate the highest attrition costs. 2. Critical Tenure: Massive spike in departures exactly at the 1-year mark. 3. Sales is the most costly department.



- *Interactive Exploration:* Drill-through capabilities allow managers to inspect individual employee profiles.
- *Custom Tooltips:* Granular data (e.g., gender distribution within roles) available on hover.



- *Transition*: Shifting from "What happened?" to "Who is at risk?".
- *The Engine*: Random Forest classification model with SMOTE balancing.
- *Business Value*: Calibrated to identify 68% of all potential leavers (prioritizing recall) using a >30% risk threshold.

```
from imblearn.over_sampling import SMOTE
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, accuracy_score

# 1. Apply SMOTE exclusively to the training data
smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)

# 2. Train the Random Forest on the balanced SMOTE dataset
rf_model_smote = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model_smote.fit(X_train_smote, y_train_smote)

# 3. Extract exact probabilities instead of strict 0/1 predictions
y_pred_probs = rf_model_smote.predict_proba(X_test)[: , 1]

# 4. Apply a Custom Decision Threshold (e.g., 0.30 instead of default 0.50)
custom_threshold = 0.30
y_pred_custom = (y_pred_probs >= custom_threshold).astype(int)

# 5. Evaluate the newly calibrated predictions
print(f"Model Accuracy (Threshold {custom_threshold}): {accuracy_score(y_test, y_pred_custom):.4f}\n")
print("Calibrated Classification Report:")
print(classification_report(y_test, y_pred_custom))
```

... Model Accuracy (Threshold 0.3): 0.7347

Calibrated Classification Report:					
	precision	recall	f1-score	support	
0	0.92	0.74	0.83	247	
1	0.34	0.68	0.45	47	
accuracy			0.73	294	
macro avg	0.63	0.71	0.64	294	
weighted avg	0.83	0.73	0.77	294	

- Model scores active employees.
- Generates an Excel report of "High-Risk Profiles".
- HR initiates proactive retention interviews.

	A	B	C	D	E	F	G
1	Attrition_Risk_Score	Age	Gender	Department	JobRole	MonthlyIncome	OverTime
2	0,85	18	Female	Sales	Sales Representative	1200	No
3	0,79	19	Male	Sales	Sales Representative	2121	Yes
4	0,76	24	Female	Sales	Sales Representative	2033	No
5	0,76	25	Male	Sales	Sales Representative	1118	Yes
6	0,76	21	Female	Sales	Sales Representative	2174	Yes
7	0,73	38	Male	Sales	Sales Representative	2858	No
8	0,72	29	Female	Research & Development	Research Scientist	2404	Yes
9	0,65	28	Male	Sales	Sales Representative	1052	No
10	0,65	27	Female	Research & Development	Research Scientist	2318	No
11	0,62	22	Female	Research & Development	Laboratory Technician	2871	No
12	0,61	21	Male	Sales	Sales Representative	3447	No
13	0,59	41	Female	Sales	Sales Representative	3140	Yes
14	0,59	41	Male	Research & Development	Research Scientist	2107	No
15	0,59	31	Male	Research & Development	Research Scientist	2321	Yes
16	0,58	27	Female	Research & Development	Research Scientist	2478	Yes
17	0,58	37	Female	Research & Development	Healthcare Representative	4777	No
18	0,58	34	Male	Research & Development	Laboratory Technician	2008	No
19	0,56	33	Male	Research & Development	Research Scientist	3348	Yes
20	0,55	18	Female	Research & Development	Research Scientist	1514	No
21	0,55	32	Female	Sales	Sales Representative	2827	No
22	0,53	40	Male	Sales	Sales Executive	9705	No
23	0,52	22	Male	Research & Development	Laboratory Technician	2935	Yes
24	0,52	37	Male	Human Resources	Human Resources	4071	No
25	0,52	31	Female	Sales	Sales Representative	2789	No
26	0,51	24	Female	Sales	Sales Executive	4577	No
27	0,5	29	Male	Research & Development	Research Scientist	2389	Yes
28	0,5	23	Female	Research & Development	Laboratory Technician	3295	No
29	0,5	24	Male	Research & Development	Laboratory Technician	2127	No
30	0,5	50	Male	Sales	Sales Representative	2683	Yes
31	0,49	29	Female	Sales	Sales Representative	2800	Yes
32	0,49	27	Female	Research & Development	Laboratory Technician	2341	No
33	0,49	28	Male	Research & Development	Healthcare Representative	8722	No
34	0,49	21	Female	Research & Development	Research Scientist	2693	No
35	0,48	29	Female	Research & Development	Research Scientist	2109	No
36	0,48	44	Male	Research & Development	Laboratory Technician	3708	No
37	0,46	40	Male	Sales	Sales Executive	9094	Yes